Extravehicular Activity Prebreathe Protocol Modeling for Exploration Missions

1.0 Objective

NASA has a requirement to produce a model that allows for the accurate prediction of Spaceflight Decompression Sickness (DCS) risk based upon evidence from research studies and operational data sources. Using retrospective data, the goal is for the model to be used as an operationally-feasible, mission-relevant DCS risk prediction tool for lunar and Mars exploration. NASA will retain ownership of data & rights of the developed model.

The contractor shall detail the modeling approach with supporting rationale describing the plans to execute the functions described in this PWS, explaining assumptions, associated risks, and provide sufficient information to demonstrate full understanding of the requirements.

2. 0 Specifications

- 2.1 The model must predict the risk of both Type I and Type II DCS and may include but is not limited to contributing factors such as variations in cabin and spacesuit pressures, sex, cardiorespiratory fitness (VO_2peak), body composition, hydration status, diet, ambulation and other activities during EVA, temperature, intermittent and continuous prebreathe protocols, and repeated exposures across multiple days of EVA. Importantly, the model will be used to assess risk during lunar and Martian surface operations of which ambulation in the spacesuit is central to the successful performance of EVA tasks that include load carrying and deployment of scientific hardware.
    - 2.1.1 The contractor may also may supplement with publicly-available or other datasets appropriate to the spaceflight DCS risk. Non-public datasets must be available to NASA at the completion of the project.
- 2.2 Models must be compliant with NASA's Standard for Models and Simulations (NASA-STD-7009) (NASA-STD-7009B-Final-3-5-2024.pdf), which includes detailed documentation of the model parameters, assumptions, and processes so that the model can be employed in operational conditions.
    - 2.2.1 The contractor should quantify the confidence and uncertainty of the risk prediction for key subgroupings of crew (e.g., sex, age, fitness) and mission-relevant conditions (e.g., cabin pressure, suit pressure, number and frequency of EVAs), if appropriate.
- 2.3 The contractor will develop the proposed model within 8-month period based upon the training dataset that NASA makes available to the funded team(s). NASA will coordinate with the contractor to complete the NASA export control process to allow the release of data to the contractor.
- 2.4 NASA will hold a subset of the entire NASA dataset for the purposes of model validation. Prediction accuracy and precision will be assessed in a masked manner by NASA statistical and modeling experts.

3.0 Required Deliverables

1) A DCS prediction tool in a compiled and executable form as well as all raw source code, spreadsheets, analysis, and configuration files needed to update, alter, and recompile the tool into a directly operable form with supporting documentation.
2) A real-time test for validation purposes. of the prediction tool to NASA using inputs, conditions, and parameters relevant to the crew, cabin environment, suit design, prebreathe conditions, and EVA on the surface of the Moon.
3) A final report detailing the methods, results (with the appropriate statistical analysis), and discussion of the findings. The report also should include:
    a. User guide for running the model
    b. Description of all datasets used to train the model, including any synthetic datasets used, rationale for approach on synthetic datasets and how those synthetic datasets were tested
    c. Validate that the model is compliant with NASA's Standard for Models and Simulations (NASA-STD-7009) (NASA-STD-7009B-Final-3-5-2024.pdf)
    d. Limitations of the model
    e. A "lessons learned" section
    f. Recommendations to improve the performance of the model and other strategies to mitigate DCS risk, as applicable.
    g. If applicable, non-public datasets used in the development of the model
4) A publication plan as well as copies (PDF) of peer-reviewed publications, if appropriate

4.0 NASA Resources

NASA will provide the modeling teams with:

- NASA-managed dataset from which to derive model
- NASA point of contact (POC) to respond to questions from the research teams. All inquiries regarding this proposal and its intent or content should be handled through this POC to ensure equal access to and consistency of information among the proposal teams.
- Content of queries from all participating teams and responses by the NASA subject matter experts ("Frequently Asked Questions", or FAQ, document) through a publicly-available website and/or email distribution.

NASA will provide data from ~1500 trials for model development but will hold a subset of the entire dataset for evaluation of the proposed model. Retrospective data from NASA will include:

- Subject demographics (e.g., age, height, weight, body mass index, cardiorespiratory fitness [peak oxygen consumption])
- Details of the decompression stress (e.g., ambient breathing gas composition, starting and final ambient barometric pressure)
- Prebreathe protocol (breathing gas composition, duration, decompression rate, resting or exercise)
- Metabolic load during prebreathe, if exercise was performed

- DCS status (none, Type I, Type II; time that symptoms reported relative to the test protocol)
- Venous gas emboli maximum Spencer grade
- Posture before, during, and after prebreathe
- Duration of and timing of events during simulated EVA following prebreathe
- Metabolic rate during simulated EVA following prebreathe

5.0 Optional datasets
While not mandatory, this additional publicly-available dataset can be incorporated into the model: CBDN Public Site at https://biodyn.istdayton.com/CBDN/Login/

Proposers also may incorporate datasets from their institution into the development of their model with a detailed description of the data source provided to NASA. Proposers should ensure that their dataset does not include data that already are contained within the NASA-provided dataset.

6.0 Supplementary Operational and DCS Information
Given that extravehicular activity (EVA) suit operations are conducted in a reduced pressure environment relative to the vehicle or habitat, there is a risk of DCS during EVA in weightlessness and partial gravity during planned exploration missions. For many years, prebreathe protocols with 100% oxygen (O_2) have been performed prior to EVA by Space Shuttle and International Space Station astronauts to reduce DCS risk and are supported by the resources that may be readily re-supplied from the ground. Resources, including crew time, will be constrained during lunar stays and missions to Mars, and astronauts will be expected to perform multiple EVAs within a mission. Efficacious DCS risk mitigation activities could decrease consumable use and crew time while maintaining or reducing risk and maximizing EVA performance, thus extending the availability of resources, protecting reserves for contingency scenarios, and elevating the potential to achieve mission objectives. Manipulating vehicle or habitat and suit environments (e.g., breathing gas composition, pressure) can impact the requirements for DCS risk mitigation, and many of these environments are not yet defined, particularly for commercial providers.

Prebreathe has been performed for EVAs in weightlessness, but the Apollo missions used a 100% O_2 vehicle atmosphere and therefore did not require prebreathe in the partial gravity environment on the lunar surface. Ambulation and increased physical workload during planetary EVAs will increase DCS risk compared with operations in microgravity, meaning that existing DCS risk mitigation protocols used on ISS likely are not applicable for planetary EVA (1). Given the limited conditions in which DCS risk mitigation strategies have been studied and the limited experience of EVAs in partial gravity, current models of DCS risk are severely limited in their ability to accurately inform exploration DCS risk mitigation protocols which can substantially affect the design and operation of vehicles, habitats, EVA suits, and launch entry and abort (LEA) suits.

7.0 Background
The goal of NASA's approach for EVA DCS risk mitigation for current and future missions is to avoid Type II (serious) DCS and to limit the incidence of Type I DCS to within acceptable limits

(NASA-STD-3001, V2 6008) or lower. The acceptable incidence of Type I DCS for a particular mission will be determined primarily based on the consequences of losing one or more EVAs due to Type I DCS. Mitigation of DCS risk typically involves trading impacts to vehicle atmospheres, spacesuit pressures (affecting crewmember dexterity and fatigue), crew time (i.e., prebreathe duration), crew hypoxia (NASA does not accept less 2.46 psia P_IO_2), and operational complexity (e.g., prebreathe masks, campout protocols).

In the Apollo program, NASA dictated the conditions in the vehicles and suits, using 100% O_2 to eliminate DCS risk and crew time requirements for prebreathe prior to EVA at the expense of increased flammability risk. NASA does not anticipate operating at vehicle atmospheres exceeding 34% O_2 due to flammability and O_2 toxicity concerns, and atmospheres involving inert gases other than N_2 have not traded favorably due to vehicle mass and complexity impacts for life support systems. The trade space is complicated by the broad array of launch, transit, and orbiting vehicles, pressurized rovers, habitats, LEA suits, and EVA suits that will enable the Moon-to-Mars architecture, with many elements being provided by international and commercial partners and the potential for a variety of potential atmospheres and operating pressures.

The majority of data used in the construction of current DCS models resulted from studies in which the baseline condition that simulated the cabin or vehicle environment was similar to Earth's sea level condition (~21% oxygen, 14.7 psia) and a common reduced pressure condition simulating the EVA suit environment (~95% oxygen, 4.3 psia); however, NASA and their international and commercial partners may choose to utilize different environments to reduce or manage DCS risk while maximizing suit mobility and controlling for flammability concerns and other hazards. Given the conditions that have been tested previously were targeted for specific purposes, decisions regarding the acceptability of cabin, habitat, rover, and suit environments that might be proposed by commercial providers and international partners are dependent upon improved modeling with a larger, more robust data set encompassing the range of environmental conditions in which an astronaut might be expected to live and work.

DCS testing protocols are developed to provide a well-defined set of conditions from which data can be extracted for models, but performance of DCS risk mitigation activities in an operational context may not be as tightly controlled or possible. For example, breaks in prebreathe protocol execution have occurred but the consequence of these and the potential mediation strategies are not well understood (2, 3). Additionally, DCS modeling to date has focused on predicting risk in a single exposure, but in the current lunar and Mars mission architecture astronauts may be expected to perform multiple EVAs in a relatively short period of time (up to 24 hours per week per astronaut which would be enabled by multiple EVAs, potentially on back-to-back days with 24 hours separating start times) though the effect of cumulative exposures is not well characterized. Further, prebreathe activities have predominantly been used for studying or conducting EVAs in weightlessness, but astronauts will perform ambulatory EVAs in partial gravity during exploration missions (1). Exercise during prebreathe is one approach that has been used to accelerate denitrogenation (4) but other methods (e.g., pharmaceuticals) have not been fully examined. Importantly, current models are based largely upon sample of male subjects and thus may not be representative of the astronaut population (5, 6).

Additionally, planning for exploration missions beyond low Earth orbit must consider contingency scenarios in the event of unplanned depressurization of the vehicle's cabin, the habitat, or the EVA suit. In the event of cabin or habitat depressurization, the pressure and duration for which the life support system must "feed the leak" to allow astronauts to don LEA spacesuits, as well as the pressure of those suits, are of critical importance. Suit pressure must also mitigate DCS risk to an acceptable level, and possibly contribute to hyperbaric DCS treatment, but it may also be necessary to transiently decrease suit pressure to improve mobility while performing certain EVA tasks to recover from the depressurization.

A summary of the current evidence for these risks, and a reference list, is available at https://humanresearchroadmap.nasa.gov/evidence/. The content relevant for this solicitation can be found under the Risk of Injury and Compromised Performance Due to EVA Operations.

The Evidence Reports can be found at:
RISK OF REDUCED CREW HEALTH AND PERFORMANCE DUE TO DECOMPRESSION SICKNESS AND HYPOXIA

RISK OF INJURY AND COMPROMISED PERFORMANCE DUE TO EVA OPERATIONS

Additionally, the Office of the Chief Medical Officer released a technical brief relevant to this topic:
Decompression Sickness (DCS) Risk Mitigation OCHMO-TB-037 Rev C

Relevant historical work in this research area was summarized in a NASA technical report:
Preventing Decompression Sickness Over Three Decades of Extravehicular Activity

Prior modeling work at NASA is summarized in a NASA technical report:
Probability of Decompression Sickness and Venous Gas Emboli from 49 NASA Hypobaric Chamber Tests with Reference to Exploration Atmosphere

Additional information can be found in:
Abercromby A, et al. Modeling a 15-min extravehicular activity prebreathe protocol using NASA's exploration atmosphere (56.5 kPa/34% O2). Acta Astronautica, Vol 109, 2015, 76-87.
https://www.sciencedirect.com/science/article/pii/S0094576514004937

REFERENCES

1. **Conkin J**, **Pollock NW**, **Natoli MJ**, **Martina SD**, **Wessel JH**, **Gernhardt ML**. Venous Gas Emboli and Ambulation at 4.3 psia. *Aerosp Med Hum Perform* 88: 370–376, 2017. doi: 10.3357/AMHP.4733.2017.

2. **Conkin J**. Decompression sickness after air break in prebreathe described with a survival model. *Aviat Space Environ Med* 82: 589–598, 2011. doi: 10.3357/asem.3020.2011.

3. **Conkin J**, **Abercromby AFJ**, **Dervay JP**, **Feiveson AH**, **Gernhardt ML**, **Norcross JR**, **Ploutz-Snyder R**, **Wessel JH**. Hypobaric Decompression Sickness Treatment Model. *Aerosp Med Hum Perform* 86: 508–517, 2015. doi: 10.3357/AMHP.4178.2015.

4. **Foster PP**, **Pollock NW**, **Conkin J**, **Dervay JP**, **Caillot N**, **Chhikara RS**, **Vann RD**, **Butler BD**, **Gernhardt ML**. Protective mechanisms in hypobaric decompression. *Aviat Space Environ Med* 84: 212–225, 2013. doi: 10.3357/asem.3314.2013.

5. **Platts SH**, **Bairey Merz CN**, **Barr Y**, **Fu Q**, **Gulati M**, **Hughson R**, **Levine BD**, **Mehran R**, **Stachenfeld N**, **Wenger NK**. Effects of Sex and Gender on Adaptation to Space: Cardiovascular Alterations. *J Womens Health (Larchmt)* 23: 950–955, 2014. doi: 10.1089/jwh.2014.4912.

6. **Strock N**, **Rivas E**, **Goebel KM**. The Effects of Space Flight and Microgravity Exposure on Female Astronaut Health and Performance. In: *2023 IEEE Aerospace Conference*. 2023 IEEE Aerospace Conference, p. 01–12.

Below are links to an additional source of data that you might consider proposing to use. The U.S. Navy DCS database has 649 saturation diving exposures that may augment altitude DCS modeling due to similarities between saturation diving conditions and ascents from sea level but proposal teams should consider the limitations. Non-saturation diving data can be considered but is may be less applicable to spaceflight exposures.

https://apps.dtic.mil/sti/pdfs/ADA451460.pdf

https://apps.dtic.mil/sti/tr/pdf/ADA452742.pdf